

A Four-Level Inverse-Problem Case Study

MATLAB implementation for a synthetic microbial consortium

Technical implementation report accompanying the manuscript

A Hierarchical Framework for Inverse Problems in Biological Cybernetics

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Abstract

This report documents a reproducible MATLAB case study illustrating four increasingly ambitious inverse problems in biological cybernetics: parameter estimation, model discovery, inverse optimal control, and inverse differential games. A common biological system is used throughout: two microbial strains that compete for a supplied substrate while reciprocally exchanging essential metabolites. The model has five dynamic states and two metabolic-allocation controls. Synthetic experiments are generated from known parameters and objectives, allowing the accuracy and limitations of each inverse step to be quantified.

The six calibrated kinetic parameters are recovered with a root-mean-square (RMS) relative error of 0.572%. Sparse model discovery obtains $R^2 = 0.950$ and $R^2 = 0.946$ for the two per-capita growth laws, while also revealing a structural ambiguity for the second strain. Inverse optimal control recovers a centralized community objective with Euclidean weight error 1.57×10^{-2} . The inverse differential-game formulation recovers the two player-specific objectives with errors 4.13×10^{-2} and 6.68×10^{-3} , and reproduces the equilibrium controls with RMS error 1.02×10^{-3} . The complete workflow executes in approximately 13 seconds on the development system, passes all four unit tests, and produces no MATLAB Code Analyzer findings.

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1 Purpose and scope

The manuscript proposes a hierarchy in which each inverse problem inherits the uncertainties of the levels below it and introduces a qualitatively new source of non-uniqueness. A useful case study should therefore do more than present four unrelated algorithms. It should hold the biological setting fixed while changing what is assumed known and what is inferred.

The present implementation was designed around five requirements:

1. use the same microbial system at all four levels;
2. keep the dynamics small enough for transparent mathematical analysis;
3. introduce explicit, biologically interpretable control variables for Levels 3 and 4;
4. provide known ground truth through synthetic data; and
5. expose non-identifiability rather than forcing exact recovery.

The result is a proof-of-concept, not a claim that objectives can already be reliably inferred from ordinary microbial abundance data. In particular, dynamic allocation measurements or appropriate molecular proxies would be needed before applying the upper levels to a real consortium.

2 Biological system and dynamic model

2.1 Two-strain cross-feeding consortium

Two microbial strains X_1 and X_2 compete for a common substrate S . Strain 1 secretes metabolite M_1 , which supports growth of strain 2, while strain 2 secretes metabolite M_2 , which supports growth of strain 1. The state is

$$\mathbf{x}(t) = [X_1(t) \ X_2(t) \ S(t) \ M_1(t) \ M_2(t)]^\top. \quad (1)$$

Each strain controls a metabolic allocation $u_i(t) \in [0, 1]$. Increasing u_i increases secretion of M_i , but reduces the strain's instantaneous growth capacity. This construction creates a biologically interpretable trade-off between individual growth and support of the partner.

2.2 Growth and material-balance equations

For strain i , let $j = 3 - i$ denote its partner. The unburdened specific growth rate is

$$\bar{\mu}_i(\mathbf{x}) = \mu_{i,\max} \frac{S}{K_{S,i} + S} \frac{M_j}{K_{M,i} + M_j}, \quad (2)$$

and the allocation-dependent growth rate is

$$\mu_i(\mathbf{x}, u_i) = \bar{\mu}_i(\mathbf{x})(1 - c_i u_i). \quad (3)$$

The implementation enforces a small positive lower bound on the allocation factor for numerical robustness. The complete dynamics are

$$\dot{X}_i = (\mu_i(\mathbf{x}, u_i) - D) X_i, \quad (4)$$

$$\dot{S} = D(S_{\text{in}} - S) - \sum_{i=1}^2 \frac{\mu_i(\mathbf{x}, u_i) X_i}{Y_{S,i}}, \quad (5)$$

$$\dot{M}_1 = \alpha_1 u_1 X_1 - \frac{\mu_2(\mathbf{x}, u_2) X_2}{Y_{M,2}} - D M_1, \quad (6)$$

$$\dot{M}_2 = \alpha_2 u_2 X_2 - \frac{\mu_1(\mathbf{x}, u_1) X_1}{Y_{M,1}} - D M_2. \quad (7)$$

Table 1 lists the ground-truth parameters used to generate the synthetic data.

Table 1: Ground-truth parameters of the synthetic microbial consortium.

Quantity	Strain 1	Strain 2
Maximum growth rate, $\mu_{i,\max}$	0.55	0.50
Substrate half-saturation, $K_{S,i}$	0.80	1.00
Metabolite half-saturation, $K_{M,i}$	0.12	0.10
Allocation cost, c_i	0.35	0.30
Secretion rate, α_i	0.55	0.50
Substrate yield, $Y_{S,i}$	0.65	0.60
Metabolite yield, $Y_{M,i}$	8.0	8.0
Dilution rate, D	0.05	
Inlet substrate, S_{in}	10.0	

3 Synthetic experiments and numerical implementation

3.1 Calibration experiments

Three calibration experiments use different initial biomass ratios, substrate levels, and metabolite concentrations. The controls are persistently exciting square-wave schedules with small sinusoidal components. They span approximately 0.08 to 0.82, which is important for separating secretion-rate effects from allocation costs. States are sampled at 49 times over 24 time units and corrupted with independent Gaussian noise at 0.5% of the maximum state scale.

3.2 Two simulation modes

The MATLAB implementation contains two complementary integrators:

- `simulateCommunity.m` uses `ode15s` with interpolated controls and non-negativity constraints;
- `simulateControlledModel.m` uses fixed-step fourth-order Runge–Kutta integration for the repeated simulations required by optimization.

The unit tests verify that the two simulators agree to an all-state RMS difference below 2×10^{-3} on a representative experiment.

3.3 Software entry point

The entire workflow is launched by

```
results = runInverseLadderCaseStudy;
```

The command generates data, executes the four levels, performs forward validation, saves `inverse_ladder_results.mat`, and exports the summary graphics.

4 Level 1: parameter estimation

4.1 Inverse problem

The model structure in equations (4)–(7) is assumed known. Six quantities are estimated:

$$\boldsymbol{\theta} = [\mu_{1,\max}, \mu_{2,\max}, \alpha_1, \alpha_2, c_1, c_2]^T. \quad (8)$$

All remaining parameters are treated as independently characterized. The weighted least-squares problem is

$$\hat{\theta} = \arg \min_{\theta_{\min} \leq \theta \leq \theta_{\max}} \sum_{e,k,\ell} \left[\frac{x_{e,k,\ell}^{\text{sim}}(\theta) - x_{e,k,\ell}^{\text{obs}}}{s_\ell} \right]^2, \quad (9)$$

where e , k , and ℓ index experiments, times, and states, respectively, and s_ℓ is a state-specific scale.

The problem is solved with `lsqnonlin` from two dispersed initial points. Bound constraints enforce biologically meaningful values.

4.2 Results

Table 2: Level 1 parameter-estimation results.

Parameter	True	Estimated	Relative error (%)
$\mu_{1,\max}$	0.55000	0.55080	0.145
$\mu_{2,\max}$	0.50000	0.49919	-0.162
α_1	0.55000	0.54964	-0.065
α_2	0.50000	0.50039	0.078
c_1	0.35000	0.35175	0.501
c_2	0.30000	0.29614	-1.287

The RMS relative error across all six parameters is 0.572%. Figure 1 shows the observed and fitted trajectories for the first calibration experiment. The fitted model reproduces biomass, substrate, and metabolite dynamics while being driven by the measured allocation schedules.

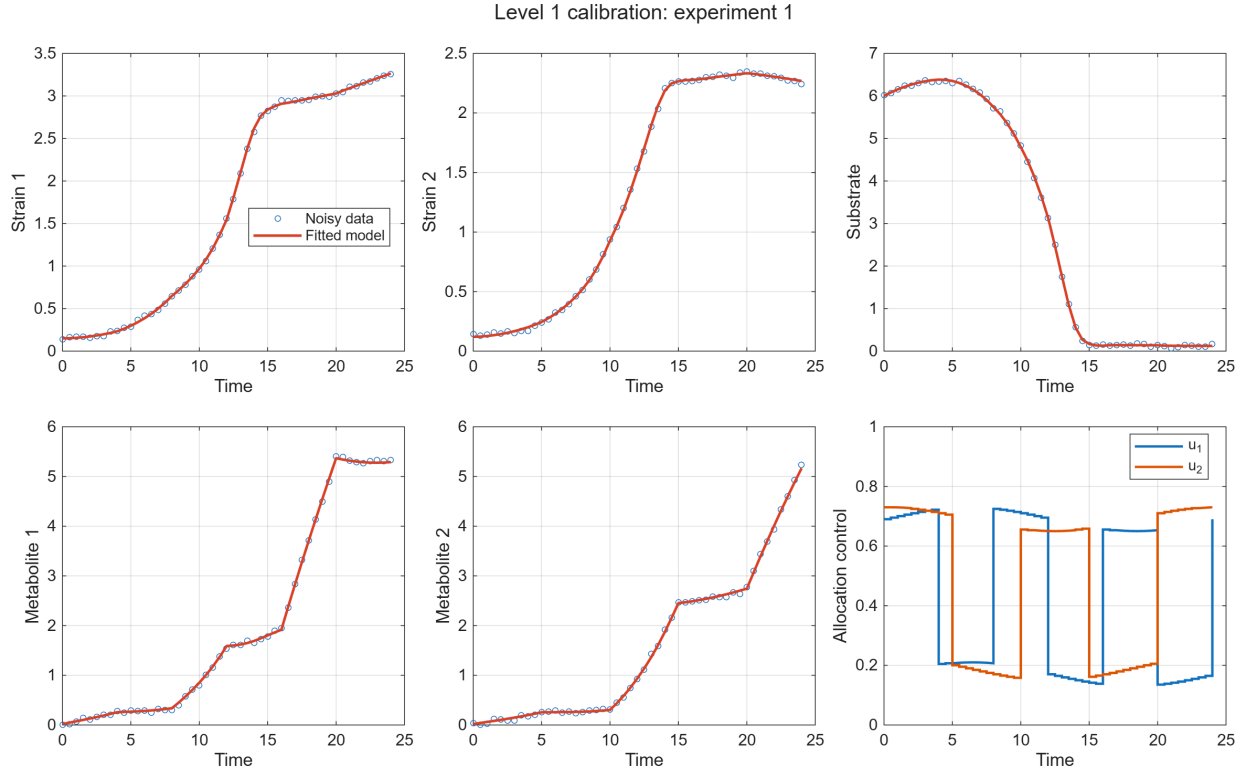


Figure 1: Level 1 calibration for the first synthetic experiment. Circles are noisy observations; solid curves are predictions obtained with the estimated parameters. The bottom-right panel shows the externally prescribed allocation controls.

5 Level 2: model discovery

5.1 Candidate library

The environmental balances are retained, while the per-capita biomass growth law is treated as unknown. For each strain, a sparse regression library is constructed from:

$$\phi = [q_i \quad q_i u_i \quad X_1 \quad X_2 \quad S \quad M_j \quad u_i], \quad q_i = \frac{S}{K_{S,i} + S} \frac{M_j}{K_{M,i} + M_j}. \quad (10)$$

Rather than differentiating noisy biomass measurements directly, the method uses window-integrated per-capita growth:

$$\frac{\log X_i(t_b) - \log X_i(t_a)}{t_b - t_a} \approx \beta_{i,0} + \frac{1}{t_b - t_a} \int_{t_a}^{t_b} \phi(t)^\top \beta_i dt. \quad (11)$$

Savitzky–Golay smoothing is applied before integration. LASSO selects a sparse coefficient vector using cross-validation.

5.2 Results and structural ambiguity

Table 3: Level 2 sparse growth-law coefficients. Zeros denote terms removed by regularization.

Candidate term	Strain 1	Strain 2
Intercept	-0.0529	-0.0252
Monod cross-feeding, q_i	0.5200	0.4157
Allocation interaction, $q_i u_i$	-0.2090	0
Strain 1 abundance	0	0
Strain 2 abundance	0	0
Substrate	0	0
Received metabolite	0	0
Direct allocation, u_i	0	-0.0496
R^2	0.9497	0.9458

For strain 1, the discovered support agrees with the generating model: growth depends on q_1 and the interaction $q_1 u_1$. For strain 2, the method selects a direct allocation term instead of $q_2 u_2$, despite achieving $R^2 = 0.946$. This is not treated as a software failure. It is an example of structural indistinguishability caused by correlated candidate functions over the observed region of state-control space. A high predictive score does not guarantee recovery of the correct biological mechanism.

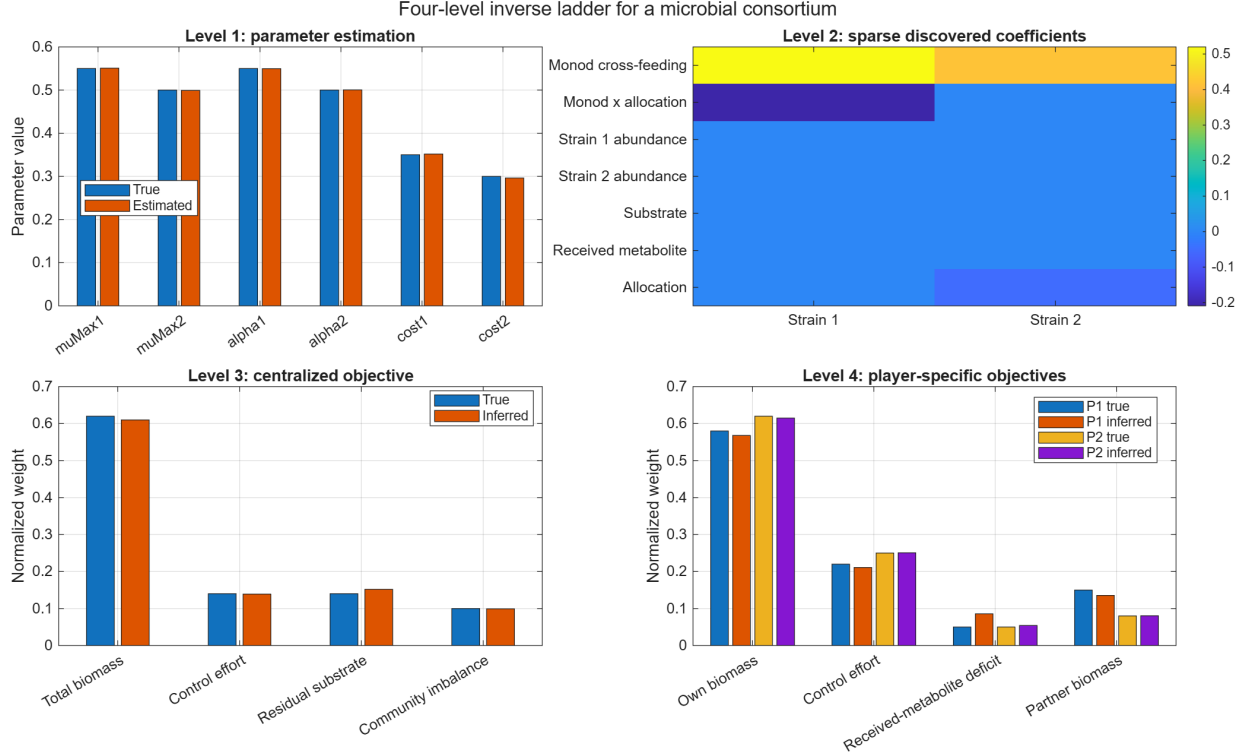


Figure 2: Summary of the four inverse levels. The Level 2 heat map displays the selected sparse coefficients, while the other panels compare ground-truth and estimated parameters or objective weights.

6 Level 3: inverse optimal control

6.1 Centralized-planner hypothesis

At Level 3, the consortium is provisionally treated as a single decision maker choosing both allocation trajectories. This is a hypothesis to be tested, not an assertion of community-level selection. The finite-horizon objective is linear in four normalized kernels:

$$J_C(\mathbf{u}; \mathbf{w}) = \sum_{m=1}^4 w_m F_m(\mathbf{x}, \mathbf{u}), \quad \mathbf{w} \geq 0, \quad \mathbf{1}^\top \mathbf{w} = 1. \quad (12)$$

The kernels represent:

1. negative total biomass, so minimization favors biomass production;
2. quadratic allocation effort;
3. residual substrate;
4. squared biomass imbalance $(X_1 - X_2)^2$.

The forward optimal-control problem uses piecewise-constant controls on eight intervals and fourth-order Runge-Kutta integration. It is solved with `fmincon` and box constraints $0.02 \leq u_i \leq 0.95$.

6.2 Inverse stationarity formulation

Because the objective is linear in $\mathbf{w}$, its control gradient at an observed optimum $\mathbf{u}^*$ can be written

$$\nabla_{\mathbf{u}} J_C(\mathbf{u}^*; \mathbf{w}) = [\nabla_{\mathbf{u}} F_1 \quad \cdots \quad \nabla_{\mathbf{u}} F_4] \mathbf{w} = A_C \mathbf{w}. \quad (13)$$

Finite differences provide the feature gradients. Rows associated with active control bounds are excluded when enough interior controls are available. The inverse problem is

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \geq 0, \mathbf{1}^\top \mathbf{w} = 1} \|A_C \mathbf{w}\|_2^2 + \lambda \|\mathbf{w} - \mathbf{w}_0\|_2^2, \quad (14)$$

with a very small regularization parameter $\lambda = 10^{-8}$ and uniform reference $\mathbf{w}_0$.

6.3 Results

Table 4: Level 3 centralized objective weights.

Objective kernel	True	Inferred
Total biomass	0.6200	0.6098
Control effort	0.1400	0.1391
Residual substrate	0.1400	0.1519
Community imbalance	0.1000	0.0991

The Euclidean weight error is 1.57×10^{-2} . Forward solution using the inferred weights returns the same discretized controls and states to the reported numerical precision. The singular values of the stacked stationarity matrix are approximately

$$8.23 \times 10^{-1}, \quad 2.31 \times 10^{-1}, \quad 2.2 \times 10^{-3}, \quad \text{near zero}.$$

The separation between the two smallest singular values is approximately 641, supporting a well-defined one-dimensional near-null direction after normalization.

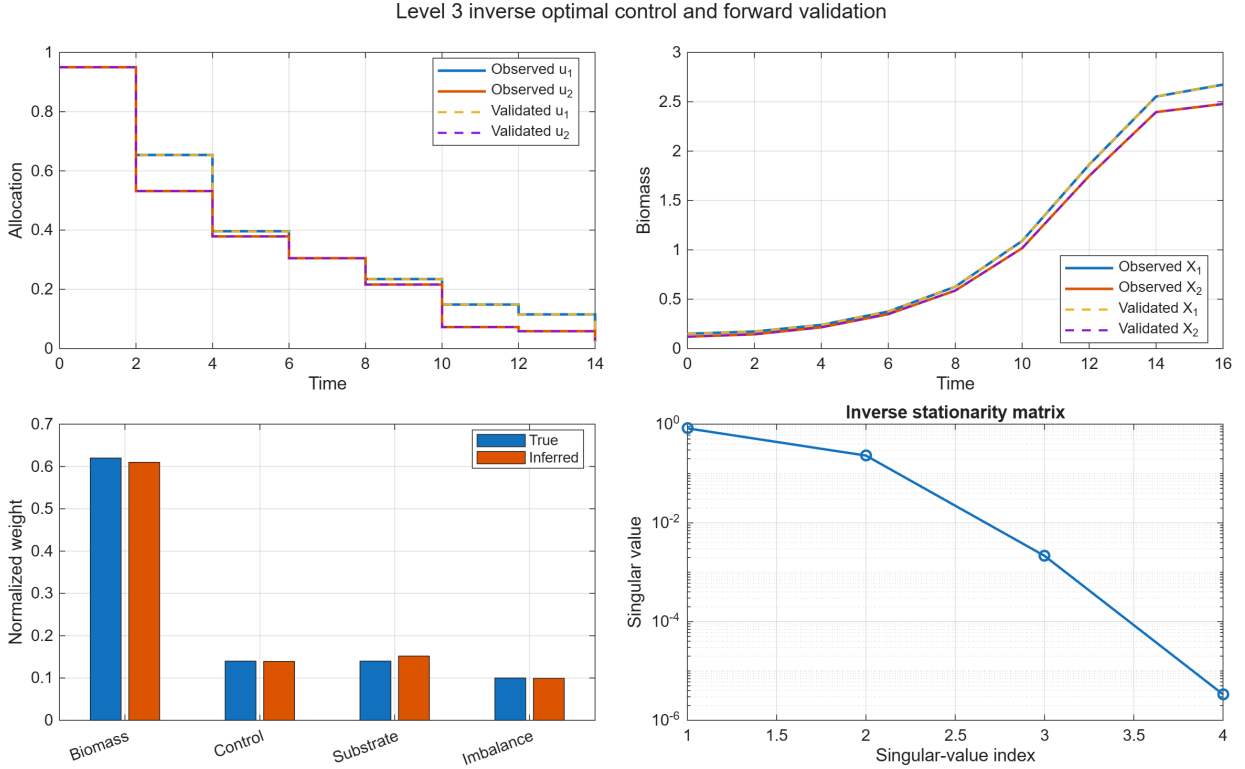


Figure 3: Level 3 results for the first experiment. The inferred objective reproduces the observed optimal controls and biomass trajectories. The bottom-right panel shows the singular-value spectrum of the inverse stationarity matrix.

7 Level 4: inverse differential game

7.1 Player-specific objectives

The centralized hypothesis is replaced by two independent decision makers. Player i minimizes

$$J_i(u_i, u_j; \mathbf{w}_i) = \sum_{m=1}^4 w_{i,m} F_{i,m}, \quad \mathbf{w}_i \geq 0, \quad \mathbf{1}^\top \mathbf{w}_i = 1. \quad (15)$$

The four kernels are:

1. negative own biomass;
2. own quadratic allocation effort;
3. deficit of the metabolite received from the partner;
4. negative partner biomass.

The final term permits a general-sum interaction: supporting the partner can be beneficial because the partner supplies an essential metabolite.

An open-loop Nash equilibrium (u_1^*, u_2^*) satisfies

$$J_i(u_i^*, u_j^*) \leq J_i(u_i, u_j^*) \quad \text{for every feasible unilateral deviation } u_i. \quad (16)$$

7.2 Forward and inverse algorithms

The forward game is solved by damped iterative best response:

1. fix u_2 and optimize u_1 ;
2. fix the updated u_1 and optimize u_2 ;
3. damp the updates and repeat;
4. verify that neither player obtains a material unilateral improvement.

Each best response is an `fmincon` problem. The inverse problem is formed separately for each player by differentiating its four feature functionals with respect to its own control:

$$\hat{\mathbf{w}}_i = \arg \min_{\mathbf{w}_i \geq 0, \mathbf{1}^\top \mathbf{w}_i = 1} \|A_i \mathbf{w}_i\|_2^2 + \lambda \|\mathbf{w}_i - \mathbf{w}_0\|_2^2. \quad (17)$$

7.3 Results

Table 5: Level 4 player-specific objective weights.

Objective kernel	Player 1		Player 2	
	True	Inferred	True	Inferred
Own biomass	0.5800	0.5679	0.6200	0.6148
Control effort	0.2200	0.2110	0.2500	0.2506
Metabolite deficit	0.0500	0.0856	0.0500	0.0541
Partner biomass	0.1500	0.1355	0.0800	0.0804

The Euclidean weight errors are 4.13×10^{-2} and 6.68×10^{-3} for Players 1 and 2. Forward validation gives:

- mean control RMS error 1.02×10^{-3} ;
- mean state RMS error 9.32×10^{-4} ;
- maximum post-solution unilateral improvement below 7.4×10^{-6} .

The near-null-direction separation ratios are approximately 20.2 and 64.0 for the two players. Player 1 is less well-conditioned, consistent with its larger weight error, particularly for the received-metabolite term.

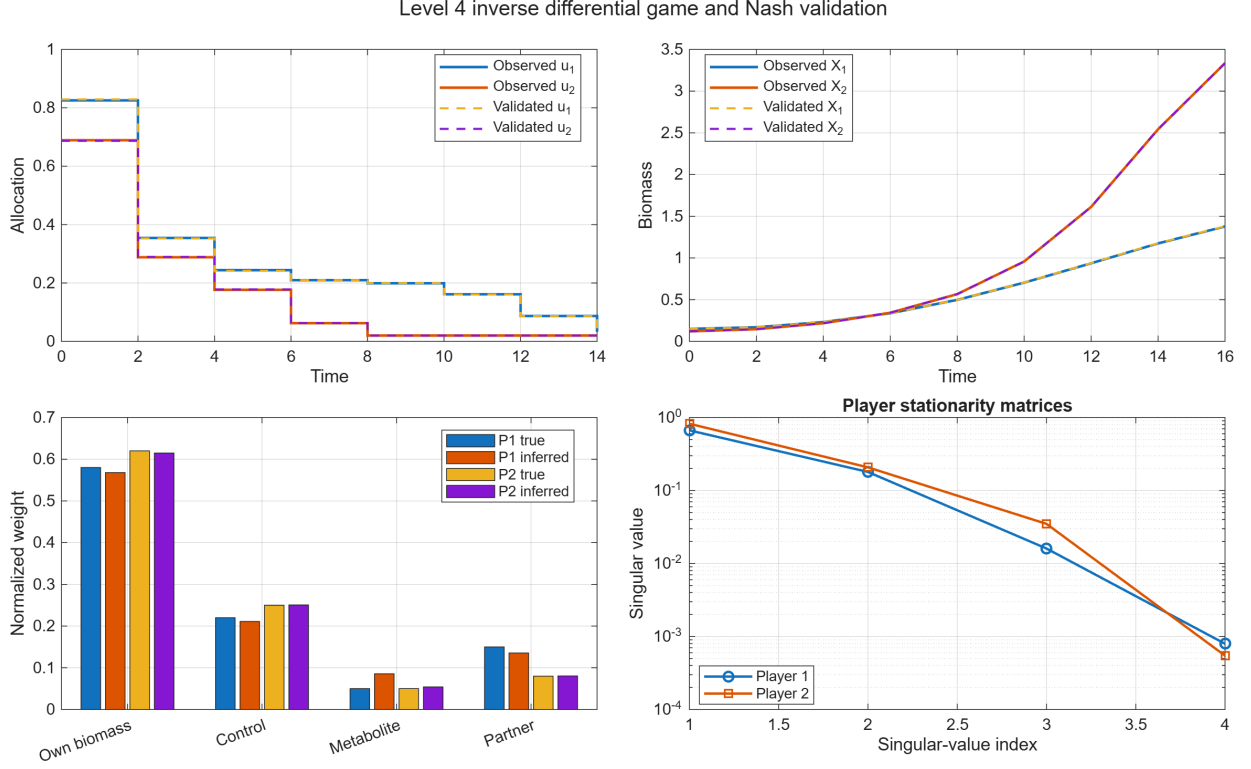


Figure 4: Level 4 results for the first experiment. Dashed forward-validation controls and trajectories closely track the equilibrium data. The lower panels compare true and inferred objectives and show the two stationarity spectra.

8 Implementation map and verification

Table 6: Principal MATLAB components.

File	Purpose
<code>runInverseLadderCaseStudy.m</code>	Top-level reproducible workflow.
<code>defaultInverseLadderConfig.m</code>	Parameters, experiments, objectives, and solver settings.
<code>communityRhs.m</code>	Five-state controlled microbial ODE.
<code>simulateCommunity.m</code>	Adaptive <code>ode15s</code> simulator.
<code>simulateControlledModel.m</code>	Fixed-step RK4 simulator used in optimization.
<code>runLevel1ParameterEstimation.m</code>	Bound-constrained multi-experiment calibration.
<code>runLevel2ModelDiscovery.m</code>	Integrated sparse growth-law discovery.
<code>solveCommunityPlanner.m</code>	Centralized forward optimal-control solver.
<code>runLevel3InverseOptimalControl.m</code>	Level 3 inverse stationarity and validation.
<code>solveOpenLoopNash.m</code>	Damped iterative best-response solver.
<code>runLevel4InverseDifferentialGame.m</code>	Player-wise inverse stationarity and Nash validation.
<code>testInverseLadderCaseStudy.m</code>	MATLAB unit tests.

The verified development environment is MATLAB R2026a Update 1 with Optimization Toolbox, Global Optimization Toolbox, Statistics and Machine Learning Toolbox, Signal Processing Toolbox, Symbolic Math Toolbox, Parallel Computing Toolbox, and Control System Toolbox. The current implementation uses the optimization, statistics, and signal-processing functionality directly.

Table 7: Verification summary.

Check	Result
Complete workflow runtime without plotting	approximately 13 s
MATLAB unit tests	4 of 4 passed
MATLAB Code Analyzer findings	0
Saved numerical result file	verified
Saved summary figure	verified
Nash unilateral-improvement check	$< 7.4 \times 10^{-6}$

9 Interpretation and limitations

9.1 What the case study demonstrates

The case study provides a computationally coherent realization of the inverse ladder:

- Level 1 recovers continuous kinetic parameters when the structure and controls are known.
- Level 2 admits competing structural explanations even when prediction is accurate.
- Level 3 recovers an objective only after choosing candidate kernels, normalizing their scale, and assuming centralized optimality.
- Level 4 requires an equilibrium concept, player-specific actions, and separate stationarity conditions.

The increasing data and modeling requirements are visible in the code. Levels 1 and 2 use ordinary time-series observations and prescribed inputs. Levels 3 and 4 additionally require controls, a candidate objective basis, and a forward optimal-control or game solver.

9.2 Important limitations

1. **Synthetic ground truth.** The trajectories, objectives, and controls are generated by the same model class used for inference. This is appropriate for verification but does not establish biological validity.
2. **Observed controls.** Real microbial studies rarely measure metabolic allocations directly. Transcript, protein, or secretion measurements would be required as proxies.
3. **Objective basis dependence.** Objective recovery is conditional on the selected feature library. An omitted kernel can redistribute inferred weights without visibly degrading the fit.
4. **Open-loop equilibrium.** Level 4 infers finite-horizon open-loop strategies. General nonlinear feedback Nash equilibria would require solving coupled Hamilton–Jacobi equations and are not attempted.
5. **Two-player scale.** The best-response and finite-difference methods are suitable for a compact illustration, not yet for large microbial communities.
6. **Local optimality information.** Inverse stationarity establishes compatibility with first-order conditions. Forward re-optimization and unilateral-deviation tests are therefore essential.

10 Recommended extensions

The most informative next developments are:

1. design additional asymmetric perturbations specifically to resolve the Level 2 allocation ambiguity;
2. add measurement noise to the Level 3 and Level 4 controls and quantify confidence regions for the inferred weights;
3. compare the centralized and game formulations on the same behavior to demonstrate cooperative mimicry explicitly;
4. replace synthetic allocations with gene-expression or metabolite-flux measurements from an engineered two-strain consortium;
5. add profile-based or Bayesian uncertainty quantification for objective weights;
6. test whether inferred objectives predict behavior under dilution, nutrient, or partner-knockdown conditions excluded from inference.

11 Conclusions

The MATLAB implementation shows that all four levels can be represented in a single biological and computational framework. Parameter estimation and sparse model discovery are inexpensive and mature enough for routine application. Inverse optimal control is feasible for a small controlled community when allocation trajectories are available. A two-player inverse differential game is

also computationally feasible under open-loop, finite-horizon assumptions, but it introduces a substantially larger modeling commitment and more severe conditioning issues.

The central lesson is not that the upper levels produce uniquely correct biological objectives. Rather, the implementation demonstrates how to formulate those hypotheses, diagnose their identifiability, and test their predictive consequences. The Level 2 structural ambiguity and the Level 4 conditioning difference between players are therefore substantive results: they make the case study consistent with the manuscript’s emphasis on both opportunities and limitations.

References

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